Infants and Incantations Design Document

Background

The idea for Infants and Incantations first came from a joke during a discussion about the story of a different game. I joked that we should make the game like *Kindergarten* and have the main character be a wizard baby that is running through a daycare causing havoc. Of course, during the meeting, we all laughed and moved onto other things. But, afterwards I found this idea sticking with me for some reason. The idea of a wizard baby crawling around and casting spells was, strangely, very comically appealing to me. As with all ideas that capture my attention, my brain pulled me inexorably towards the inevitable conclusion–making the game myself.

As I developed the story more, I moved away from the initial idea of a gory but comical game like *Kindergarten* into an action RPG that had more of what I enjoy about video games: high player customisability, in depth mechanics, and unique enemy designs. Eventually, the game settled into what it is now: A 3D, real-time, action-RPG where the player is a wizard baby who is attempting to escape a demonically run daycare.

The game is still in the prototype phase. So, this document will continue to be updated regularly, as well as it being subject to change.

Overview

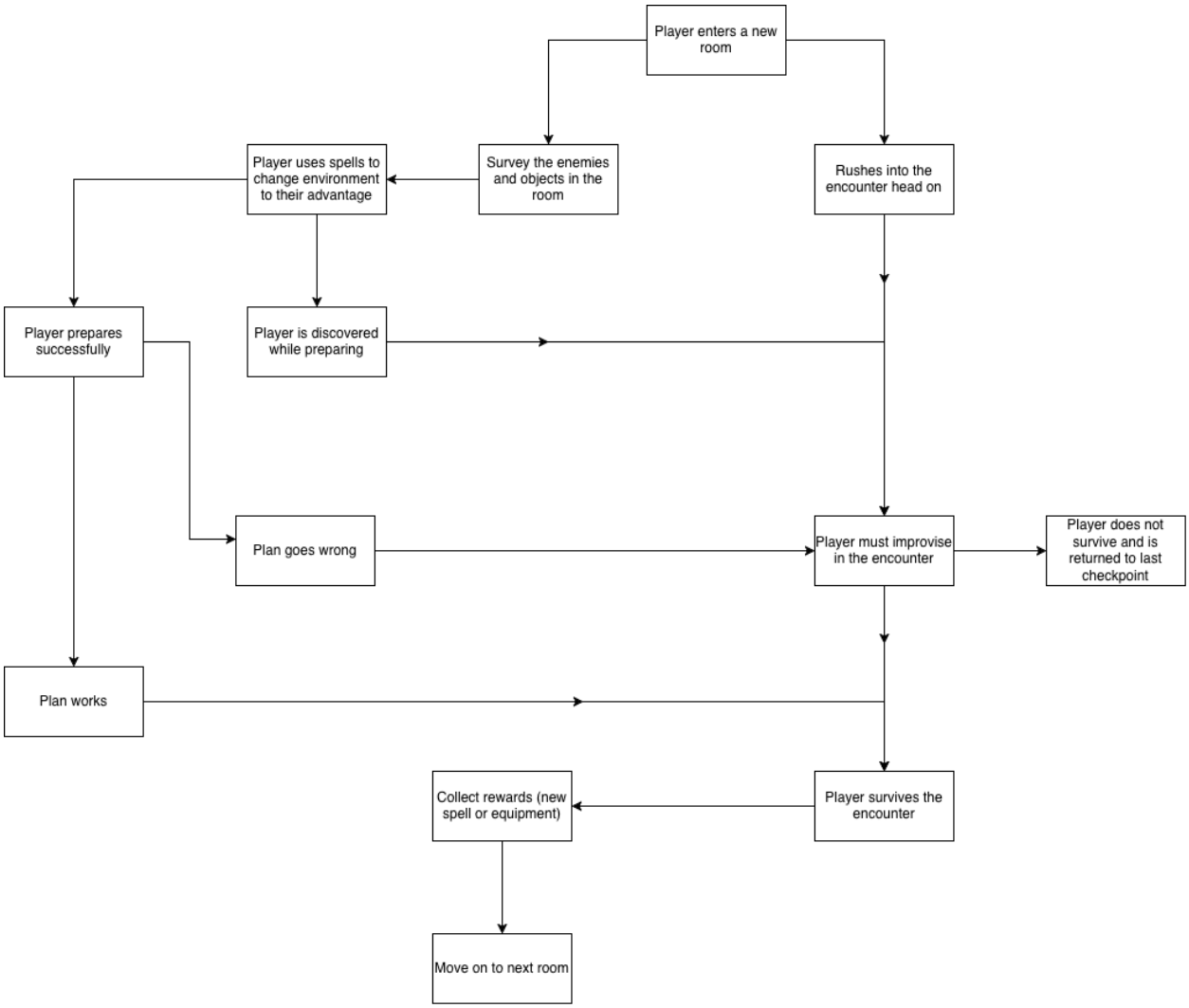
Infants and Incantations is a game where the player is a spellcasting baby attempting to break out of a daycare that is being run by demons. It is a 3D game that uses an isometric camera perspective[1] and WASD movement, spells will be aimed and cast using the mouse. The player will battle different enemy types as the story continues, which will get more powerful as they progress. The player will have an arsenal of spells that they can equip a small subset of. These spells will be able to manipulate the environment to the advantage of the player in combat, and defeat enemies using damage.

The core gameplay loop will be the player exploring rooms of the daycare to find more and better spells or equipment. In each room, there will either be an encounter or a puzzle. Encounters will not begin until the player is noticed by an enemy, either by walking into their line of sight or by attacking them. Enemies will also investigate obvious commotion created by the player in the room. For example, if a player casts a spell when they are close to an enemy; when a projectile passes close by an enemy; or when an object in the environment moves significantly. When investigating something, they will search around the area of the disturbance, and if they find the player it will begin the combat encounter. Once started, an encounter won't end until the player has either won by defeating all of the enemies present in the room, or lost by being defeated themself (health reduced to 0).

[1] However, I am heavily considering changing it to be in third person due to it potentially simplifying aiming of spells and/or movement for the player. Once the prototype is built, this will be one of the things I test.

Combat and puzzles are centered around the use of spells to manipulate the environment. In combat encounters, the player will use their spells to move objects in the room to set up an advantageous position for the encounter. This can occur both before and during an encounter. The puzzles will be focused on encouraging the player to use their spells in ways that they had not considered, or teaching them how to use a new spell (like shrines in Breath of the Wild).

The gameplay flow for combat encounters should look like this:



Every few three rooms (± 1 or 2 rooms) there will be a checkpoint location. These will behave very similar to bonfires from the Dark Souls series and will restore the player's health to its maximum, provide the player with an opportunity to edit their loadout of spells and equipment, allow fast travel from checkpoint to checkpoint, and restore all encounters to their original state (reviving defeated enemies and restoring interactable objects). As it does not occur every room, it rewards careful play over multiple rooms, as well as giving some punishment for dying, as the player will have to go back through encounters they may have already finished.

Story

Synopsis

The player is playing as a baby, who possesses innate magical and sorcerous abilities. The game begins just after the baby's parents drop them off at what they believe to be a perfectly normal daycare, then leave to go to work. However, unknown to the parents, the daycare has been taken control of by demons, whose only goal is to get filthy rich by using the powers of magical babies like the player for themselves.

The story will follow the player as they attempt to leave the daycare, and must battle upwards through the bureaucracy to eventually find and confront the demonic shareholders of the daycare. At first, the player character is attempting to leave for purely selfish reasons and is unaware of the daycare's corruption. However, as they continue upwards through the bureaucracy the player character becomes aware of the corruption, as well as the positive impact that their actions are having on the other babies trapped in the daycare, and they will grow to be fighting not just for themself but for all of the other babies trapped in the daycare.

Progression

The player will first take control of the player character when being put in time out. They are brought into a room by an imp disguised as a daycare attendant, and put in a cot. This will function as a kind of tutorial scene for the player, where they will defeat the imp and be able to leave the time out room.

After leaving the time out room, the player emerges onto the first floor which is where all of the babies of the same age as the player character. These babies are watched over and prevented from leaving by more of the imps disguised as daycare attendants. The player defeats these imps and tries to leave through the front door. However, they find the front door is sealed by a ward that only allows passage if you file the correct form with the floor 1 manager. The player is then forced to scour each classroom on the floor to find the classroom manager. Eventually, the player finds the manager's office and enters it, beginning the first real boss fight of the game. After defeating the manager, they are subdued and forced to sign the form that allows the player to leave. But, the manager is not allowed to sign the form without permission from the floor 2 manager.

The player must then go in the elevator to the next floor, where the imps are combined with devils (also disguised as attendants). As the player explores this floor trying to find the office of the floor manager, they help more of the other babies trapped in this daycare. From these babies, they hear that the shareholders have a prophetic baby trapped in the board room. They also hear that the baby is being forced to use its powers of prophecy to predict how the stock market will behave each day so that the shareholders can make money from daytrading. Troubled by this, but not necessarily deterred from their goal, the player continues to find the manager of the floor and defeat them in another boss fight. After another victory, the player will try and file their form that they acquired from the previous manager. This manager will also not be able to and will direct the player up to the board room and shareholders, as the

player character is technically considered an employee of the company and requires the shareholders to release him from employment.

The player character is extremely aggravated by this information and declares that they will make the shareholders release all of the babies from the daycare. So, they go in the elevator to the next floor which is dedicated to the shareholders and daytraders that they employ (all of which are imps, devils, and demons). This is where the shareholders' plan to profit off of the prophetic baby is revealed to the player. The player battles through the offices and eventually makes it to the board room, where the shareholders are meeting. A boss fight ensues, and the player emerges victorious. They force the demons to release the ward, allowing all of the babies to leave the daycare freely. They also free the prophetic baby and undo the anchor that was allowing the demonic daycare attendants to manifest in the world, freeing each of the babies. The player then leaves the building with all of the babies, just in time for the parents to all arrive to pick them up, ending the game.

Game Mechanics

Guiding Principles

I wanted the mechanics at play to represent the difference in physicality between the player character and the enemies. The player is a baby, and is much more physically weak than the enemies are. Therefore, the combat should be approached carefully, and the player should have to outwit the enemy.

To support the above design philosophy, I want the player to be able to shape the environments during, or before, an encounter. The spells are then primarily for this purpose. The player should have to set up for an encounter ahead of time, preparing the environment in some way to allow their currently equipped spells to be best utilised.

This creates important considerations for level or environment design, as I need each encounter to allow the player to be–at least slightly–prepared for it. The player also needs some way of actually preparing the environment for the encounter ahead. For example, setting up boxes in a specific pattern to maximise the damage of a ricochet spell.

However, to avoid the player being able to "optimize the fun out of the game" the player should be discouraged from reusing the same setup for each encounter. This introduces a few challenges when designing the encounters, as the environment and enemy designs need to encourage the player to experiment and improvise in new scenarios. This could also be done by introducing new enemy types that also interact with the environment.

I must also ensure that not too much weight is placed on the pre-planning aspect of each encounter. While planning out strategies and preparing before a fight is fun, it is often a lot more fun when having to improvise when a plan falls apart, or accidentally walk into an encounter without thinking and still somehow make it out alive. So, when a plan falls apart, or an interaction doesn't work out the way the player thinks it will, it shouldn't be a death sentence. Instead, there should be a secondary improvisational aspect to the encounter where the player has to utilise the environment to their advantage on the fly. For example, by pushing enemies into each other and creating a chain reaction of stuns to allow the player to pick them off one at a time.

Overall, the player should feel powerful through clever utilisations of magic and the environment, not through their physicality.

We are left with these overall principles to this end:

1. Spells should enable or reward planning for an encounter and manipulate the environment in advance of it (spells can fill both roles at the same time).
2. Each encounter should have opportunities for the player to plan ahead.
3. Every room that could potentially be a part of an encounter should have parts that can be moved or rearranged by the player.

4. Encounters need to be designed to be varied and encourage the player to experiment.
5. The player should be allowed to improvise in the encounters.

Spells

As stated above, there should be two main categories of spells: spells that encourage the manipulation of the environment to reach their full effect, and spells that allow the manipulation of the environment. These spells should be or discovered as the player progresses throughout the game to allow more customisation of loadout and playstyle. To avoid spamming a spell during an encounter, when the player is in an encounter a cooldown will be placed on a spell after casting it. This cooldown will be decided on a per-spell basis, with more powerful spells having a longer cooldown.

The current spells I have are as follows:

- Push spell
    - A spell that pushes all objects away from the player..
    - The spell AOE is a small cone that faces the direction of the player's cursor.
    - It charges up while the player holds the cast button, increasing the length of the cone up to a limit of 1 second (5 meters).
    - If an object is pushed into an enemy by this spell: the object is destroyed and the enemy takes damage that scales with the speed (velocity * mass) that the object was moving at, and is briefly stunned.
        - Creates an interesting trade off between needing the object for later in the encounter, but needing a short stun now to get away.
    - Tentative cooldown: 1 second.
- Ricochet spell
    - Fires a beam in the direction of the player's cursor.
    - The beam deals damage to enemies, but deals more damage the more times it ricochets off of a surface–to a maximum of 4 times (5, 15, 25, 40).
    - The beam cannot damage the same enemy multiple times, but can pierce through enemies.
        - Instead of firing a beam through one enemy repeatedly for 4 bounces, incentivise finding angles that hit multiple enemies, or one enemy after bounding for 4 times.
    - After travelling for 30 meters, or when it hits the fifth surface, it dissipates.
    - Tentative cooldown: 5 seconds.
- Firebolt spell
    - Fires a small ball of fire in the direction of the player's cursor.
    - The projectile grows in size the further it travels, capping out at 30 meters (1m diameter).
    - The projectile does a very small amount of damage if it hits an enemy directly but, if it hits a non-enemy, it will create an explosion that increases in size and damage proportionally to the size of the projectile.
        - AOE radius (meters): projectile diameter * 5
        - AOE damage: projectile diameter * 30
    - Tentative cooldown: 5 seconds.

Numbers and formulae for calculating numbers are subject to change after playtesting and feedback. More spells will also be added as the game grows.

Equipment

Armour and accessories are outside the scope of the current prototype, but I eventually want to introduce equipment that the player can use to further customise their playstyle. While I personally enjoy the freedom that is offered by having many accessory and armour slots (Terraria), I don't think having many slots would suit this game. Firstly, the direction the design is going is towards encouraging the player to make the most out of limited resources in a situation where they are outmatched. I believe that having many slots for equipment and armour would take away from this feeling. However, this is something I am willing to experiment with after playtesting and feedback.

Enemies

Enemies will, of course, get stronger over the course of the game. But I don't want a specific type of enemy to stop appearing. If a player figures out how to counter an enemy, they should be able to use that knowledge for the remainder of the game. There will be large umbrella types of enemies that will be introduced on each floor (imps on the first floor, devils on the second, and demons on the third), but each of these types will have subtypes with different behaviors. The classification of imp/demon/devil will mainly be used to telegraph to the player how strong that enemy is.

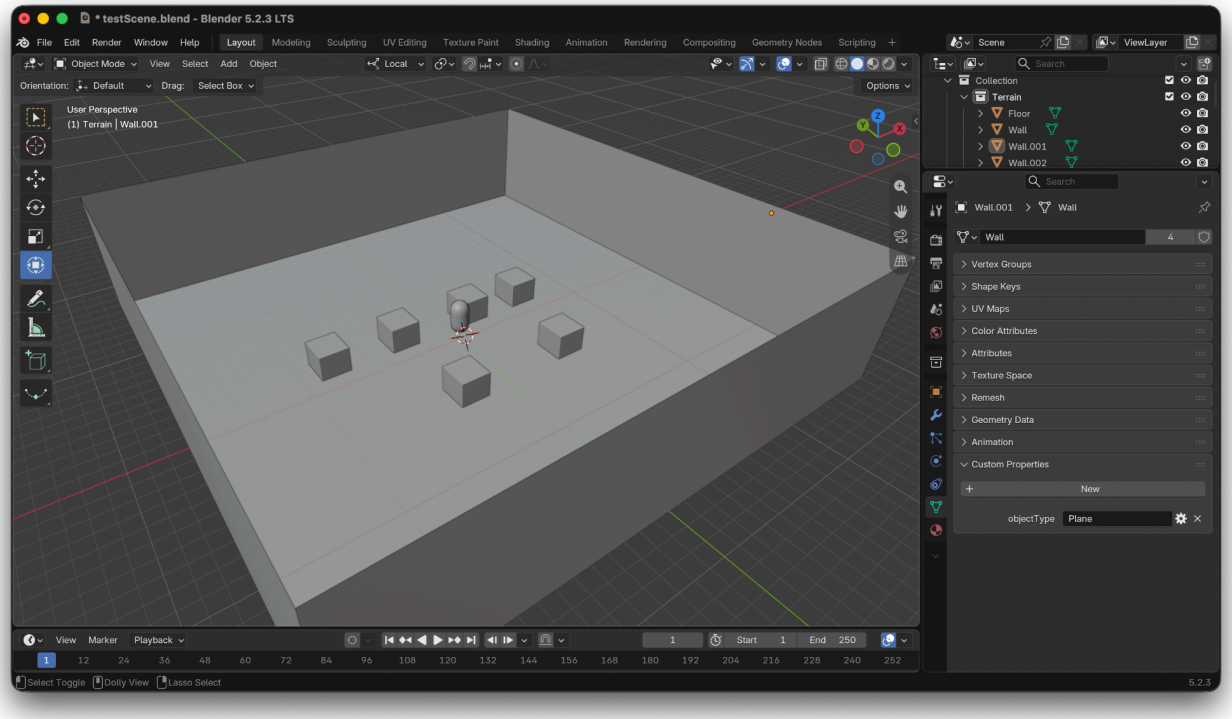
For the scope of the prototype, I am constrained by my technical knowledge of programming enemy AI's as this is something I have not done before. As a result, I am going to make the enemy present in the prototype fairly simple to ensure I learn the fundamentals and set the groundwork for developing more complex AI's later. The enemy will be an imp that is disguised as a basic adult caretaker. It will have a basic melee attack that will "grapple" the player by picking them up and returning them to the timeout cot (which will result in a game over screen and restarting the prototype). The "grapple" attack will have a very short range, and will be slow to activate in order to give the player ample time to run away. The enemy is not very complex as it is the first enemy in the game, and I don't want to drop the player into the deep end mechanically.

Prototype

Prototype development is focused on creating the technical fundamentals for the game in order to provide some kind of proof of concept. The goal is to create a single encounter with a basic enemy, an environment that the player can experiment with, and spells that are interesting to use. Overall, the prototype is intended to be a vertical slice of the combat encounters for the game (see the flow chart in the overview section).

The scope of the prototype is limited to a single room, with enough quantity and variety in objects to make the encounter interesting. The player will have access to the three spells that were detailed in the spells section.

As I test and iterate on the prototype, I will continue to update this document. But, the scene currently looks this this in blender:



My current priorities are fully implementing the functionality of the prototype. But, when it is done, I will move on to designing and creating the encounter.

The process of creating the encounter will include the creation of actual 3D meshes for each object, the player character, and the enemy–rather than placeholders; variety in the types of objects that are in the room; iterating on the layout and design of the level; and finally, reworking spell and enemy behavior and numbers to ensure balance.